INCH-POUND
MIL-DTL-83731/10E
16 September 2019
SUPERSEDING
MIL-DTL-83731/10D
28 January 2003

DETAIL SPECIFICATION SHEET

SWITCHES, TOGGLE, MINIATURE, LEVER SEAL, PANEL SEAL, DOUBLE POLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-DTL-83731.

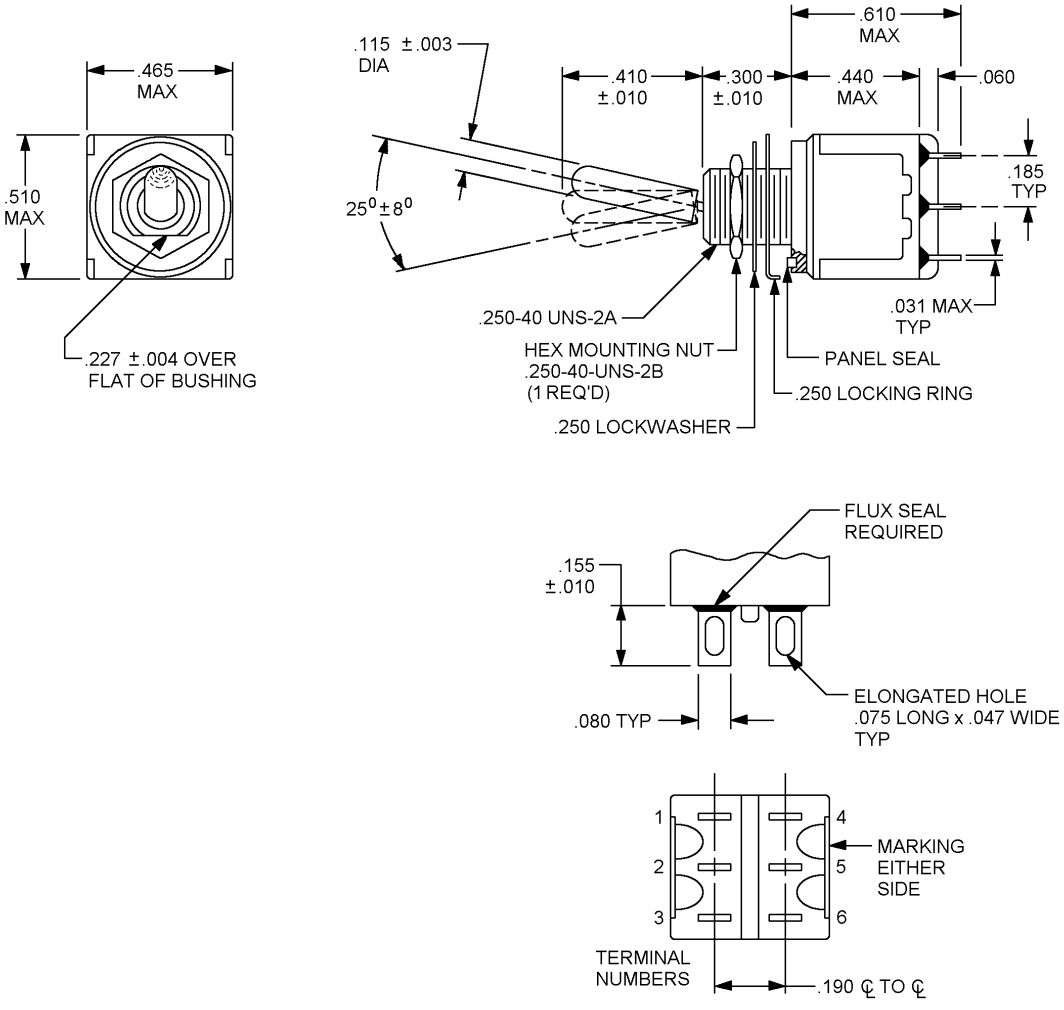


FIGURE 1. Solder lug termination.

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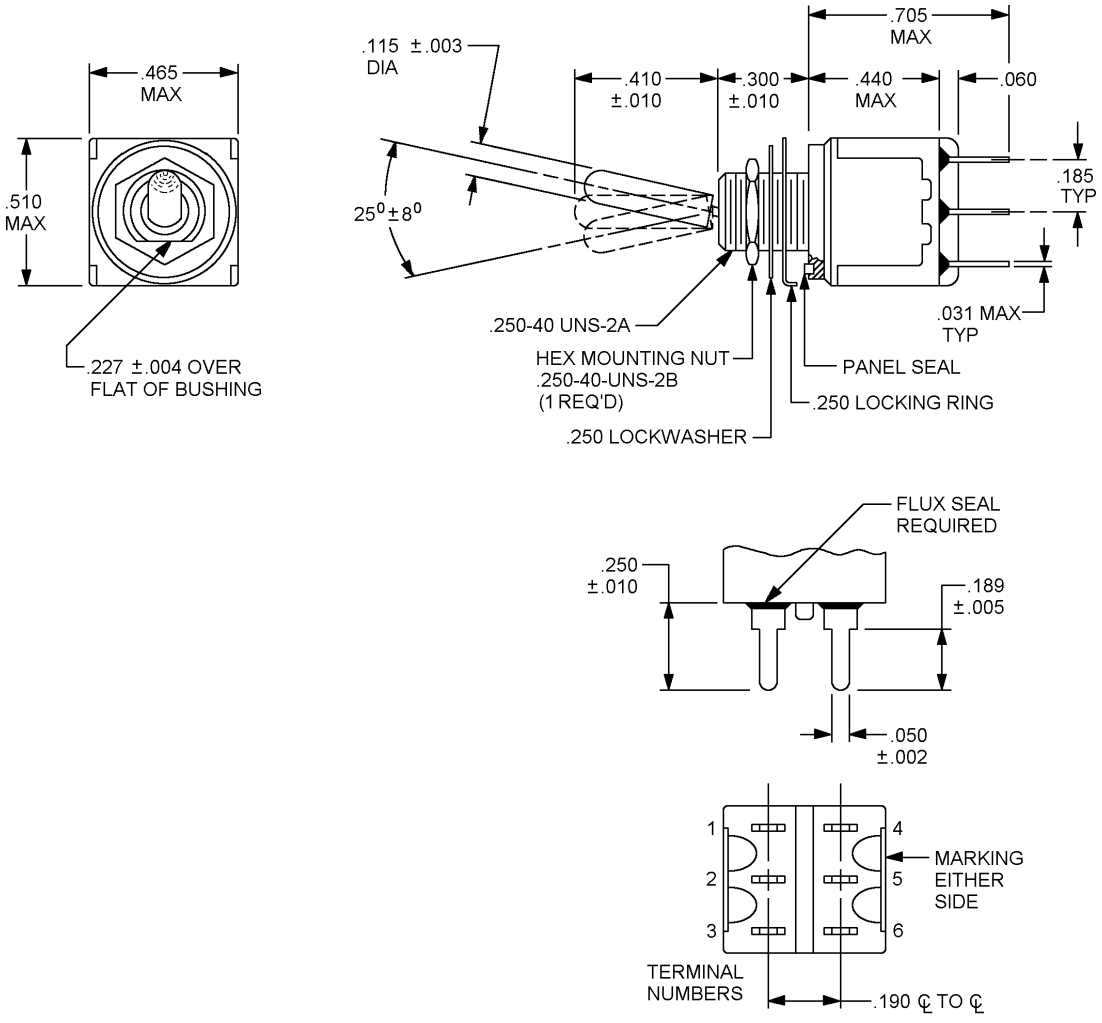
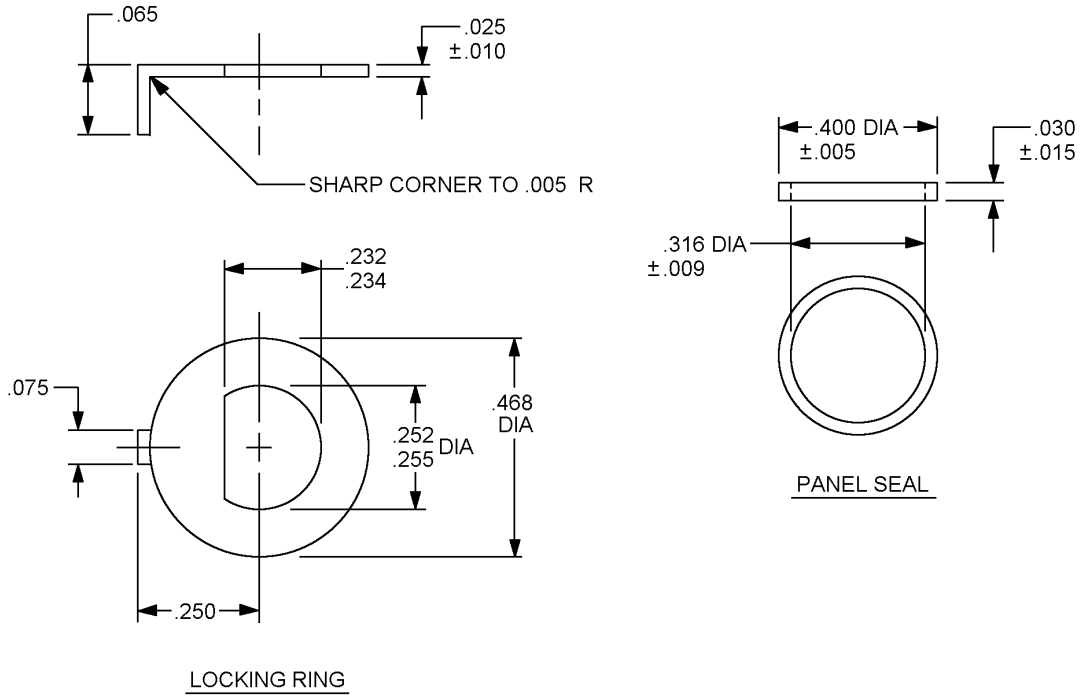


FIGURE 2. Printed circuit termination.

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NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ± .005.
3. Direction of internal mechanism movement is opposite to direction of toggle movement.
4. Mounting bushing is D-shaped.
5. To insure proper sealing by panel seal, locking ring should be used on front of panel only.

FIGURE 3. Locking ring and panel seal.

REQUIREMENTS:

Design and construction: See figures 1 and 2.

Material and finish. Bushings shall be brass, nickel plated. Toggle lever shall be brass, bright chrome plated. Frame shall be stainless steel. The contacts shall be gold plate over nickel over silver over brass.

Mounting hardware: Each switch shall be supplied with one hexagon nut, one internal-tooth lockwasher in accordance with the appendix of MIL-DTL-83731, one locking ring (see figure 3), and one panel seal (see figure 3). The panel seal shall be silicone rubber, or Buna-N.

Weight: 0.019 pound maximum.

Electrical ratings: See table I.

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Strength of terminals: MIL-STD-202-211,test condition A. The applied force shall be one pound.

Strength of toggle lever, pivot, and lever stop: Test (a) shall use a ten pound load; test (b) shall be a 8 pound load.

Dielectric withstanding voltage: Test at atmospheric pressure only.

Mechanical endurance: 10,000 cycles at -25°C +0°C, -4°C, and 10,000 cycles at 71° +4°, -0°C.

Electrical endurance, electrical overload, intermediate current, logic level life, and temperature rise. Multipole switches are to be tested with a load on each pole and with the same polarity on adjacent poles in accordance with figure 2.

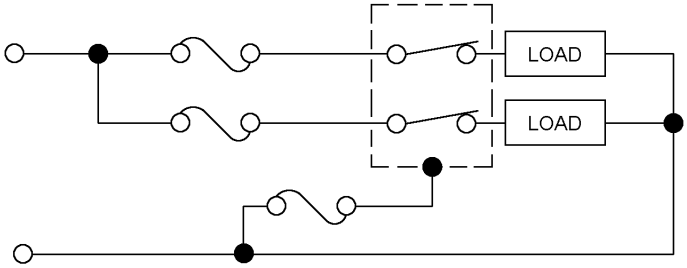


FIGURE 4. Test circuit.

TABLE I. Electrical ratings.

Type of load	High level (current in amperes)			Logic level
	28 V dc	125 V ac, 60 Hz	115 V ac, 400 Hz	5 V dc
Resistive	5	5	5	0.010 ampere
Inductive	2	2	2	- - -
Lamp	1	1	1	- - -

Resistive load (dc and ac): Switches shall make and break the rated resistive load for 10,000 operating cycles at room ambient pressure and a temperature of 71°C +4°C -0°C.

Inductive load (dc and ac): Switches shall make and break the rated inductive load for 10,000 operating cycles at room ambient temperature and pressure.

Lamp load: Switches shall make and break the rated lamp load for 10,000 operating cycles.

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Logic level load: 10,000 cycles at an actuation rate of 10-18 cycles per minute with no “stick” or “misses” allowed when tested in accordance with EIA448, method 17 as follows:

Each pair of switch contacts shall be tested using a 5.0 ±0.05 V dc, 10 ± mA resistive load. During each closure of the contacts, the voltage drop across the switch terminals shall be monitored for a duration of no less than 50 percent of each contact static closure. The switch contacts need not be monitored until 10 ms after the initial contact closure to exclude any contact bounce. During each opening of the contacts, the voltage drop across the switch terminals shall be monitored for a duration of no less than 50 percent of each contact opening.

A voltage of 2.1 volts or greater across the switch terminals shall constitute a contact “miss”, (failure to properly close the circuit). A voltage drop of less than 90 percent of the open-circuit voltage shall constitute a contact “stick” (failure to properly open the circuit).

The monitoring device shall either record the contact closures at which “sticks” or “misses” occur, or discontinue the test when “sticks” or “misses” occur.

Operating temperature:

a. 50 percent of the test cycle at room ambient conditions.

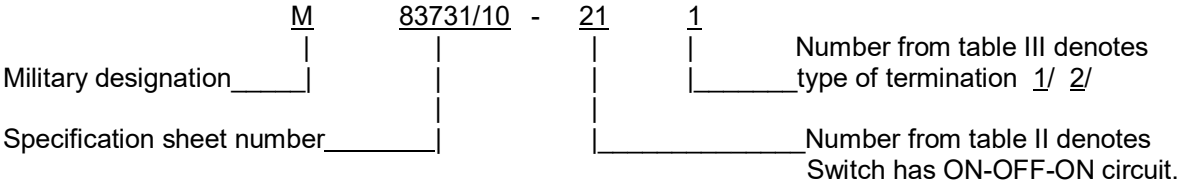
b. 50 percent of the test cycle at 71 +4°C or -0°C.

Short circuit: Use 100 amperes at 28 V dc.

Flux sealed: The sealing is obtained when the base is sealed to prevent flux from entering the switch case during the wave soldering process.

Flux seal test: Three switches shall be tested as follows during qualification and group B inspection: Measure and record initial contact resistance. Place switches, terminals down, into a shallow pan. Pour flux at 80°F ±5°F, type RMA of J-STD-004, specific gravity 0.896, into pan without splashing until level of flux is approximately .062 inch (1.57 mm) above the bottom of the plastic switch case and let switches soak for ten minutes. Remove switches from flux, clean with flux cleaning solvent and immediately put into oven for drying at 175°F for two hours. After switches have cooled at room temperature, repeat initial measurement. Contact resistance shall not increase by more than 10 milliohms over the initial reading. Disassemble and visually examine the contact area for evidence of flux shall be cause for rejection.

Military Part or Identifying Number (PIN): The PIN shall consist of the prefix M83731/10 followed in order by the appropriate two-digit number from table II and the number of table III as shown in the following example:



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TABLE II. Circuit for switching characteristic.

Circuit	Circuit with toggle in		
	Toward flat (down position)	Center	Opposite flat (up position)
21	1-2 ON 4-5	OFF	2-3 ON 5-6
23	1-2 ON 4-5	NONE	2-3 ON 5-6
26	1-2 MOM-ON 4-5	NONE	2-3 ON 5-6
27	1-2 MOM-ON 4-5	OFF	2-3 MOM- ON 5-6
31	1-2 MOM-ON 4-5	OFF	2-3 ON 5-6
32	1-2 ON 4-5	2-3 ON 4-5	2-3 ON 5-6
33	1-2 MOM-ON 4-5	2-3 ON 4-5	2-3 ON 5-6
34	1-2 MOM-ON 4-5	2-3 ON 5-6	NONE
35	1-2 MOM-ON 4-5	2-3 ON 4-5	2-3 MOM-ON 5-6

TABLE III. Number for termination type.

Type of termination	
Solder lug	1
Printed circuit	2

1/ Listed dash numbers, ending with the number 1, are interchangeable with and supersede canceled dash numbers ending with the numbers 3 and 5.
2/ Listed dash numbers, ending with the number 2, are interchangeable with and supersede canceled dash numbers ending with the numbers 4 and 6.

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Referenced Documents:
MIL-DTL-83731
MIL-STD-202-211
EIA448
J-STD-004

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodians:
Army - CR
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5930-2019-032)

Review activities:
Army - AR

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